

YIHANG TAO

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EDUCATION

City University of Hong Kong

PhD, Department of Computer Science
Supervisor: Prof. Yuguang "Michael" Fang
JC STEM Lab of Smart City, WINET Laboratory

Hong Kong SAR
Sep. 2024 - Now

Shanghai Jiao Tong University

Master, School of Computer Science
Outstanding Graduate (2024)

Shanghai, China
Sep. 2021 - Apr. 2024

Southeast University

Bachelor, School of Information Engineering
GPA: 3.89/4.0

Nanjing, China
Sep. 2017 - Jul. 2021

RESEARCH INTERESTS

World Model, Generative Model, 3D Vision, Autonomous Driving.

PUBLICATIONS

First Author (* Equal contribution)

[**Under Review**] **Y. Tao**, Y. Guo, Z. Fang, H. An, and Y. Fang. "FRUC: Feedforward Dynamic Scene Reconstruction from Uncalibrated Collaborative Driving Views," *Under Review*. [[PDF Link](#)]

[**Under Review**] **Y. Tao***, Y. Guo*, S. Hu, Y. Ma, Z. Fang, S. Kwong, and Y. Fang. "V2XCrafter: Learning to Generate Driving Scene Across Agents," *Under Review*. [[PDF Link](#)]

[**CVPR'26**] **Y. Tao**, S. Hu, H. An, Z. Fang, H. Cao, and Y. Fang. "Learning Mutual View Information Graph for Adaptive Adversarial Collaborative Perception," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. [[CCF-A](#)] [[PDF Link](#)]

[**TDSC**] **Y. Tao***, S. Hu*, Y. Hu, H. An, H. Cao, and Y. Fang. "GCP: Guarded Collaborative Perception with Spatial-Temporal Aware Malicious Agent Detection," *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2026. [[CCF-A](#)] [[PDF Link](#)]

[**ICRA'25**] **Y. Tao**, S. Hu, Z. Fang, and Y. Fang. "Directed-CP: Directed Collaborative Perception for Connected and Autonomous Vehicles via Proactive Attention," *IEEE International Conference on Robotics and Automation (ICRA)*, Atlanta, USA, 2025. [[CCF-B](#)] [[PDF Link](#)]

[**AAAI'25 Oral**] S. Hu*, **Y. Tao***, G. Xu, Y. Deng, X. Chen, Y. Fang, and S. Kwong. "CP-Guard: Malicious Agent Detection and Defense in Collaborative Bird's Eye View Perception," *The 39th Annual AAAI Conference on Artificial Intelligence (AAAI)*, Philadelphia, USA, 2025. (**Oral Presentation, <5%**) [[CCF-A](#)] [[PDF Link](#)]

[**TGCN**] **Y. Tao**, J. Wu, Q. Pan, A. K. Bashir, and M. Omar. "O-RAN-Based Digital Twin Function Virtualization for Sustainable IoV Service Response: An Asynchronous Hierarchical Reinforcement Learning Approach," *IEEE Transactions on Green Communications and Networking (TGCN)*, vol. 8, no. 3, pp. 1049-1060, Sep. 2024. [[PDF Link](#)]

[**GLOBECOM'23**] **Y. Tao**, J. Wu, X. Lin, S. Mumtaz, and S. Cherkaoui. "Digital Twin and DRL-Driven Semantic Dissemination for 6G Autonomous Driving Service," *IEEE Global Communications Conference (GLOBECOM)*, Kuala Lumpur, Malaysia, Dec. 2023, pp. 2075-2080. [[CCF-C](#)] [[PDF Link](#)]

[**LNET**] **Y. Tao**, J. Wu, X. Lin, and W. Yang. “DRL-Driven Digital Twin Function Virtualization for Adaptive Service Response in 6G Networks,” *IEEE Networking Letters (LNET)*, vol. 5, no. 2, pp. 125-129, Jun. 2023. [[PDF Link](#)]

Collaborative Author

[**Under Review**] Y. Guo*, Z. Fang*, S. He, S. Hu, **Y. Tao**, P. Lin, and Y. Fang. “Universal Image Restoration via Internalized Chain-of-Thought Reasoning,” *Under Review*. [[PDF Link](#)]

[**Under Review**] Z. Fang*, Y. Guo*, F. Liu, Y. Zhang, **Y. Tao**, S. Hu, W. Ding, and Y. Fang. “Agent-Centric Visual Reinforcement Learning under Dynamic Perturbations,” *Under Review*. [[PDF Link](#)]

[**ICML’26**] S. Hu, Y. Dai, Y. Zhao, **Y. Tao**, Y. Guo, Z. Fang, S. T. W. Kwong, and Y. Fang. “Optimizing Agentic Reasoning with Retrieval via Synthetic Semantic Information Gain Reward,” *Forty-Third International Conference on Machine Learning (ICML)*, 2026. [[PDF Link](#)]

[**CVPR’26**] H. An, Y. Xiaohui, G. Hua, **Y. Tao**, H. Cao, X. Yu, and Y. Fang. “RecoverMark: Robust Watermarking for Localization and Recovery of Manipulated Faces,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. [[CCF-A](#)] [[PDF Link](#)]

[**TDSC**] H. An, G. Hua, W. Du, H. Cao, **Y. Tao**, G. Xu, S. Rahardja, and Y. Fang. “Decoder Gradient Shields: A Family of Provable and High-Fidelity Methods Against Gradient-Based Box-Free Watermark Removal,” *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2026. [[CCF-A](#)] [[PDF Link](#)]

[**TMC**] S. Hu, **Y. Tao**, G. Xu, X. Qian, Y. Deng, X. Chen, S. Kwong, and Y. Fang. “CP-uniGuard: A Unified, Probability-Agnostic, and Adaptive Framework for Malicious Agent Detection and Defense in Multi-Agent Embodied Perception Systems,” *IEEE Transactions on Mobile Computing (TMC)*, 2025. [[CCF-A](#)] [[PDF Link](#)]

[**NeurIPS’25**] Y. Guo, S. He, Y. Lu, H. An, **Y. Tao**, H. Zhu, J. Liu, and Y. Fang. “Neptune-X: Active X-to-Maritime Generation for Universal Maritime Object Detection,” *The 39th Annual Conference on Neural Information Processing Systems (NeurIPS)*, **Spotlight**, San Diego, USA, Dec. 2025. [[CCF-A](#)] [[PDF Link](#)]

[**NeurIPS’25**] S. Hu, X. Han, J. Jiang, **Y. Tao**, Z. Fang, Y. Dai, S. Kwong, and Y. Fang. “Distribution-Aligned Decoding for Efficient LLM Task Adaptation,” *The 39th Annual Conference on Neural Information Processing Systems (NeurIPS)*, San Diego, USA, Dec. 2025. [[CCF-A](#)] [[PDF Link](#)]

[**GLOBECOM’25**] S. Hu, Y. Ma, **Y. Tao**, Z. Fang, Z. Fang, Y. Deng, S. Kwong, and Y. Fang. “Task-Aware Parameter-Efficient Fine-Tuning of Large Pre-Trained Models at the Edge,” *IEEE Global Communications Conference (GLOBECOM)*, Taipei, Taiwan, Dec. 2025. [[CCF-C](#)] [[PDF Link](#)]

[**JSAC**] Z. Fang, J. Wang, Y. Ma, **Y. Tao**, Y. Deng, X. Chen, and Y. Fang. “R-ACP: Real-Time Adaptive Collaborative Perception Leveraging Robust Task-Oriented Communications,” *IEEE Journal on Selected Areas in Communications (JSAC)*, 2025. [[CCF-A](#)] [[PDF Link](#)]

[**TMC**] Z. Fang, Z. Lin, S. Hu, **Y. Tao**, Y. Deng, X. Chen, and Y. Fang. “Dynamic Uncertainty-aware Multimodal Fusion for Outdoor Health Monitoring,” *IEEE Transactions on Mobile Computing (TMC)*, 2026. [[CCF-A](#)] [[PDF Link](#)]

HONORS & AWARDS

Outstanding Academic Performance Award (OAPA), City University of Hong Kong	Aug. 2025
IEEE Robotics and Automation Society (RAS) Travel Grant for ICRA’25	Mar. 2025
Outstanding Graduate, Shanghai Jiao Tong University	May 2024
WEICHAI POWER Scholarship, Shanghai Jiao Tong University	Oct. 2023

Excellent League Member, Shanghai Jiao Tong University Apr. 2022
National Student Research Training Program Excellence Award (**Leader**) Oct. 2020
Excellence Prize, 2nd International Data Competition, IKCEST (**top 3%**) Oct. 2020
Sun Qingyun Innovation Scholarship, Southeast University (<**1% annually**) Jun. 2020
Finalist, 36th Mathematical Contest in Modeling (MCM), COMAP (**top 1%**) Apr. 2020
First Prize, 12th National Information Security Contest, China (**top 8%**) Aug. 2019
First Prize, 15th Advanced Mathematics Competition, Jiangsu (**top 10%**) Aug. 2018

ACADEMIC SERVICE

Program Committee Member:

CVPR, ECCV, ICML, NeurIPS, ICLR, ACM MM, IROS, ICASSP, IJCNN

Journal Reviewer:

IEEE TMC, IEEE TITS, IEEE TCE, TMLR, PR, NN, EAAI, IEEE JBHI, IEEE LNET

Journal Administrator:

IEEE TMLCN

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, C++, C#, JavaScript, Verilog HDL, $\LaTeX$, Markdown

Deep Learning Frameworks: PyTorch, TensorFlow, OpenMMLab (MMDetection, MMSegmentation)

Tools & Platforms: Git, Docker, Linux, CARLA Simulator, SUMO, ROS

Languages: English (IELTS 7.0), Chinese (Native)